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| Embedded System Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  | Semester    | 06  |
| Course Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | BEC601                           | CIE Marks   | 50  |
| Teaching Hours/Week (L:T:P: S)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3:0:2:0                          | SEE Marks   | 50  |
| Total Hours of Pedagogy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 40 hours Theory + 8-10 Lab slots | Total Marks | 100 |
| Credits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 04                               | Exam Hours  |     |
| Examination nature (SEE)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Theory                           |             |     |
| <b>Course objectives:</b> <ul style="list-style-type: none"><li>● Identify various components, their purpose, and their application to the embedded system's applicability.</li><li>● Program various embedded components using the embedded C program.</li><li>● Understand the embedded system's real-time operating system and its application in IoT</li><li>● Understand the fundamentals of ARM-based systems, including architecture and its units like registers , debug interface, stack, MPU, Interrupts etc</li><li>● Use the various instructions to program the ARM controller.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |             |     |
| <b>Teaching-Learning Process (General Instructions)</b> <p>These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.</p> <ol style="list-style-type: none"><li>1. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the delivered lessons shall develop students' theoretical and applied Mathematical skills.</li><li>2. Provide real-life examples.</li><li>3. Support and guide the students for self–study.</li><li>4. You will assign homework, grading assignments and quizzes, and documenting students' progress.</li><li>5. Encourage the students to group learning to improve their creative and analytical skills.</li><li>6. Show short related video lectures in the following ways:<ul style="list-style-type: none"><li>● As an introduction to new topics (pre-lecture activity).</li><li>● As a revision of topics (post-lecture activity).</li><li>● As additional examples (post-lecture activity).</li><li>● As an additional material of challenging topics (pre-and post-lecture activity).</li><li>● As a model solution of some exercises (post-lecture activity).</li></ul></li></ol> |                                  |             |     |
| <b>MODULE-1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |             |     |
| <b>Introduction to Embedded System:</b> What is an Embedded Systems? Embedded systems Vs General computing systems, History of Embedded Systems, Classification of Embedded systems, Major Application Areas of Embedded Systems. Purpose of Embedded Systems, The Typical Embedded System, Microprocessor Vs Microcontroller, Differences between RISC and CISC, Harvard V/s Von-Neumann Processor/Controller Architecture, Big-endian V/s Little-endian processors, Memory (ROM and RAM types), Sensors & Actuators, The I/O Subsystem – I/O Devices, Light Emitting Diode (LED), 7-Segment LED Display, Optocoupler, Relay, Piezo buzzer, Push button switch, Communication Interfaces, On-board Communication Interface, External Communication Interface, Embedded Firmware, Other System Components                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  |             |     |
| <b>(Text 1: All the Topics from Ch-1 and Ch-2.)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                  |             |     |
| <b>MODULE-2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |             |     |
| <b>Embedded System Design Concepts:</b> Characteristics and Quality Attributes of Embedded Systems, Operational and non-operational quality attributes, Embedded Systems-Application and Domain specific, Hardware Software Co-Design and Program Modeling (excluding UML), Embedded firmware design and development (excluding C language).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                  |             |     |

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| <b>Text 1: Ch-3, Ch-4 (4.1, 4.2.1 and 4.2.2 only), Ch-7 (Sections 7.1, 7.2 only), Ch-9 (Sections 9.1, 9.2, 9.3.1, 9.3.2 only)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>MODULE-3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>RTOS and IDE for Embedded System Design:</b> Operating System basics, Types of operating systems, Task, process and threads (Only POSIX Threads with an example program), Thread preemption, Preemptive Task scheduling techniques, Task Communication, Task synchronization issues – Racing and Deadlock. How to choose an RTOS, Integration and testing of Embedded hardware and firmware, Embedded system Development Environment – Block diagram (excluding Keil).<br><b>(Text 1: Ch-10 (Sections 10.1, 10.2, 10.3, 10.5.2, 10.7, 10.8.1.1, 10.8.1.2 only), Ch-12, Ch-13 (a block diagram before 13.1, only)).</b> |
| <b>MODULE-4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>ARM Embedded Systems:</b><br>Introduction, RISC design philosophy, ARM design philosophy, Embedded system hardware – AMBA bus protocol, ARM bus technology, Memory, Peripherals, Embedded system software – Initialization (BOOT) code, Operating System, Applications.<br>ARM Processor Fundamentals, ARM core dataflow model, registers, current program status register, Pipeline, Exceptions, Interrupts and Vector Table, Core extensions.<br><b>Text 2: Chapter 1, 2</b>                                                                                                                                         |
| <b>MODULE-5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Introduction to the ARM Instruction set:</b> Introduction, Data processing instructions, Load – Store instruction, Software interrupt instructions, Program status register instructions, Loading constants, ARMv5E extensions, Conditional Execution.<br><b>Text 2: Chapter 3</b>                                                                                                                                                                                                                                                                                                                                     |

#### PRACTICAL COMPONENT OF IPCC

Conduct the following experiments on an ARM CORTEX M3 evaluation board to learn Assembly Language Program and using evaluation version of Embedded 'C' & Keil uVision-4 tool/compiler.

| Sl.NO | Experiments                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------|
| 1     | Write a program to find the sum of the first 10 integer numbers.                                        |
| 2     | Write an Assembly Language Program (ALP) to i) Multiply two 16-bit numbers. ii) Add two 32-bit numbers. |
| 3     | Write a program to find the factorial of a number.                                                      |
| 4     | Write a program to add an array of 16 bit numbers and store the 32 bit result in internal RAM.          |
| 5     | Write a program to find the square of a number (1 to 10) using a look-up table.                         |
| 6     | Write a program to find the largest or smallest number in an array of 32 numbers.                       |
| 7     | Write a program to arrange a series of 32 bit numbers in ascending/descending order.                    |
| 8     | Write a program to count the number of ones and zeros in two consecutive memory locations.              |
| 9     | Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction.                      |
| 10    | Interface a DAC and generate Triangular and Square waveforms.                                           |
| 11    | Display the Hex digits 0 to F on a 7-segment LED interface, with a suitable delay in between.           |
| 12    | Interface a simple Switch and display its status through Relay, Buzzer and LED                          |

**Course outcomes (Course Skill Set):**

At the end of the course, the student will be able to:

- Describe the architectural features and instructions of 32-bit microcontroller ARM Cortex M3.
- Apply the knowledge gained for Programming ARM Cortex M3 for different applications.
- Understand the basic hardware components and their selection method based on the characteristics and attributes of an embedded system.
- Understand the hardware software co-design and firmware design approaches.
- Explain the need of real time operating system for embedded system applications.

**Assessment Details (both CIE and SEE)**

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

**CIE for the theory component of the IPCC (maximum marks 50)**

- IPCC means practical portion integrated with the theory of the course.
- CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.
- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other assessment methods mentioned in 22OB4.2. The first test at the end of 40-50% coverage of the syllabus and the second test after covering 85-90% of the syllabus.
- Scaled-down marks of the sum of two tests and other assessment methods will be CIE marks for the theory component of IPCC (that is for **25 marks**).
- The student has to secure 40% of 25 marks to qualify in the CIE of the theory component of IPCC.

**CIE for the practical component of the IPCC**

- **15 marks** for the conduction of the experiment and preparation of laboratory record, and **10 marks** for the test to be conducted after the completion of all the laboratory sessions.
- On completion of every experiment/program in the laboratory, the students shall be evaluated including viva-voce and marks shall be awarded on the same day.
- The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report. Each experiment report can be evaluated for 10 marks. Marks of all experiments' write-ups are added and scaled down to **15 marks**.
- The laboratory test (**duration 02/03 hours**) after completion of all the experiments shall be conducted for 50 marks and scaled down to **10 marks**.
- Scaled-down marks of write-up evaluations and tests added will be CIE marks for the laboratory component of IPCC for **25 marks**.
- The student has to secure 40% of 25 marks to qualify in the CIE of the practical component of the IPCC.

**SEE for IPCC**

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours**)

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.

4. Marks scored by the student shall be proportionally scaled down to 50 Marks  
**The theory portion of the IPCC shall be for both CIE and SEE, whereas the practical portion will have a CIE component only. Questions mentioned in the SEE paper may include questions from the practical component.**

**Suggested Learning Resources:**

**Text Books**

1. Shibu K V, "Introduction to Embedded Systems", Tata McGraw Hill Education
2. Andrew N Sloss, Dominic System and Chris Wright, "ARM System Developers Guide", Elsevier, Morgan Kaufman publisher, 1st Edition, 2008.

**Reference Book**

1. Raj Kamal, "Embedded Systems: Architecture and Programming", Tata McGraw Hill, 2008.

**Web links and Video Lectures (e-Resources):**

1. <https://archive.nptel.ac.in/courses/106/105/106105193/>
2. <https://developer.arm.com/documentation/dui0068/b/ARM-Instruction-Reference>
3. <https://www.udemy.com/course/introduction-to-arm-cortex-m3-and-m4-processors/>
4. [www.Nuvoton.com](http://www.Nuvoton.com)/websites on Advanced ARM Cortex Processors
5. <https://alison.com/tag/embedded-systems>

**Activity Based Learning (Suggested Activities in Class)/ Practical Based learning**

Programming Assignments / Mini Projects can be given to improve programming skills